Response to Reviewers

1. General Response

We thank the Associate Editor and all five reviewers for their careful and constructive evaluation. The reviews were detailed and fair, and they have substantially improved the manuscript. We are encouraged that the reviewers found the problem **important** (R1, R2, R4) and the functional-decoupling perspective **theoretically well-motivated** (R5), including gains over a strong CLIP ViT-L/14 baseline and larger improvements on hard intent categories (R1, R5).

We have organized the revision around four major concerns:

- **Fair comparison and calibration (R1).** We added a matched-feature controlled protocol, an original-pipeline robustness check, a method-versus-thresholding decomposition, multi-seed reporting, and threshold-free metrics. We use the matched-feature setting as the controlled comparison and report the original pipelines only as a complementary check.
- **Novelty and theoretical depth (R2, R4).** We now state a formal decoupling principle, test it with a unified-versus-decoupled control, and use negative controls to show that the result does not follow from arbitrary external text. An auxiliary study on an adjacent subjective dataset is presented as secondary evidence of cross-task generalization.
- **Evaluation and interpretation (R3, R4).** We added recent zero-shot MLLM baselines, ROC/PR and t-SNE visualizations, per-class attribution, efficiency comparisons, and a second-dataset study. The main manuscript reports the conclusions and controlled comparisons; detailed auxiliary results remain in tables or released artifacts.
- **Clarity and reproducibility (R1, R4, R5, EiC).** We clarified the purpose and limitations, added a notation table and Discussion, documented validation-only selection, committed to a concrete artifact release, and corrected the bibliography, "FDIR" typo, and Figure 2 caption.

Below, we provide point-by-point responses to each reviewer. Newly added or revised content is referenced by section, table, or figure. In the revised manuscript, all changes are highlighted in red.

2. Response to Reviewer 1

We thank the reviewer for a very constructive review and for recognizing the importance of the problem and the empirical interest of FDIL.

Comment 1

The main concerns are the fairness and interpretability of the comparisons. Please provide controlled baselines under the same backbone and thresholding protocol.

Response: This is an important fairness concern. We now compare FDIL, the CLIP baseline, HLEG, LabCR, and IntCLIP using the same frozen CLIP ViT-L/14 features, data split, and validation-only threshold search. Under this matched protocol, FDIL leads the three representative external methods by +3.4 **to** +4.7 **Avg F1**. We retain results from the methods' original pipelines only as a complementary robustness check and do not use their different backbones to support the FDIL claim. To make the source of the controlled gain interpretable, we also report per-class attribution: the largest improvements occur in abstract, disagreement-prone intents, while only two of the 28 classes regress.

Revised manuscript: Sec. 4, Table 2, and the Interpretability Analysis; complete per-class values in Table R1 of this letter.

Table R1: Per-class F1 attribution of FDIL versus the CLIP baseline under the matched class-wise protocol. The table reports the five largest improvements and the only two regressions among all 28 classes.

Intent class	Subset	Baseline	FDIL	Δ
WorkILike	Hard	14.6	43.0	+28.5
PassionAboutSomething	Hard	11.3	28.6	+17.3
SocialLifeFriendship	Hard	33.9	51.2	+17.2
ManageableMakePlan	Medium	17.5	32.0	+14.5
BeatCompete	Medium	54.6	68.1	+13.5
FineDesignLearnArt-Arch	Easy	84.0	83.1	−1.0
CreativeUnique	Medium	41.9	36.6	−5.3

Comment 2

The strong effect of class-wise thresholding.

Response: To separate calibration gain from method gain, we added a four-cell, three-seed comparison of the CLIP baseline and FDIL under global and class-wise thresholds. FDIL improves Avg F1 under both class-wise (+4.08) and global (+2.31) thresholding, while class-wise calibration benefits both methods. We further report four threshold-free metrics and a 5,000-resample paired bootstrap test. FDIL improves over the CLIP baseline on every threshold-free metric; its mean mAP gain is +4.43, and the representative-seed difference is +4.80 with 95% CI $[3.58, 6.10]$ and $p < 0.001$.

Revised manuscript: Sec. 4 (Calibration and Threshold-free Evaluation), Tables 9 and 10.

Comment 3

Report variance over runs.

Response: Following this suggestion, the main comparison and thresholding decomposition now report mean±std over **three seeds** for UTD-only and full FDIL.

Revised manuscript: Sec. 4, Tables 2 and 9.

Comment 4

Clarify validation-only model selection.

Response: We have made the selection boundary explicit and audited the implementation: early stopping uses validation metrics only, and all thresholds are fitted on validation and fixed before test evaluation. Test outputs are used solely for final reporting and never enter a model- or threshold-selection branch.

Revised manuscript: Sec. 4 (Experimental Setup and Implementation Details).

Comment 5

The dependence on offline VLM-generated rationales and semantic priors, and incomplete reproducibility details. Please release or document generated artifacts.

Response: We now state this dependence and its cost explicitly: rationales and class scenarios are generated offline, the UTD text teacher is discarded after training, and the full FDIL model retains the lightweight semantic-prior reranking stage. We document feature extraction, prior construction, teacher encoding, optimization, threshold search, and the agreement gate. We also replaced "available on request" with a concrete commitment to release the generated priors, rationales, cached features/logits, exact prompts, thresholds, seeds, checksums, and training/evaluation scripts upon acceptance.

Revised manuscript: Sec. 3; Sec. 4 (Implementation Details); Sec. 5 (Limitations and Future Directions); and the Data and artifact availability statement.

Comment 6

Tone down claims about functional decoupling and SOTA performance unless the ablations isolate these factors more cleanly.

Response: We agree that these claims require evidence that separates architectural organization from backbone and calibration effects. We therefore added a series of stricter experiments, including three-seed evaluation, global- and class-wise-threshold controls, threshold-free metrics with paired bootstrap tests, and unified-versus-decoupled ablations under identical features and selection rules. The most direct unified control, which merges the semantic prior and rationale teacher into one distillation target, reaches 55.66 **Avg F1**, whereas decoupled FDIL reaches 57.49; the component and negative-control studies further confirm that both the image-matched rationale and agreement-aware use of supervision contribute to the result. For a fair comparison with recent state-of-the-art approaches, we also reproduced HLEG, LabCR, and IntCLIP using the same frozen CLIP ViT-L/14 features, data split, and validation-only class-wise thresholding. FDIL retains a +3.4 **to** +4.7 **Avg F1** advantage under this matched protocol. We have accordingly removed unsupported absolute-SOTA wording and now claim that functional decoupling is an empirically validated, interpretable design with a measurable aggregate advantage under controlled comparisons.

Revised manuscript: Abstract; Sec. 4, Tables 2, 6, and 10.

3. Response to Reviewer 2

We thank the reviewer for acknowledging that the manuscript is well-written, technically sound, and empirically validated, and for raising the central novelty concern.

Comment 1

The major drawback is related to insufficient novelty of the technical contribution. Most of the contributions proposed in the paper relate to previously discussed topics within vision-language modeling, semantic alignment, and uncertainty-aware learning. Although the idea of distinguishing between semantic uncertainty and supervision uncertainty is quite appealing, it does not constitute a new theoretical principle or algorithmic idea. It appears that the results are based mainly on the correct combination of existing methods rather than the development of a novel approach itself.

Response: We acknowledge that the original presentation did not distinguish the methodological contribution clearly enough from a combination of existing components. We have therefore revised the method around a formal, testable principle: supervisory ambiguity is a label-side conditional-entropy term, whereas semantic ambiguity is a feature-side aliasing term, so their corrections act on different factors of the risk. We now provide both empirical motivation and a theoretical explanation for this principle. Empirically, the candidate-recall diagnostic shows that the baseline Top-10 set already covers 83.9% of positive labels and retains 67.8% of its false negatives as locally recoverable candidates, indicating that much of the remaining error is local ranking and calibration rather than complete retrieval failure. Theoretically, the new excess-risk decomposition separates label-side conditional entropy from feature-side inter-class aliasing and shows why the two corrections act on different objects. Together, these results motivate separate local ranking/calibration and supervision-stabilization functions rather than a post-hoc architectural split.

We test this principle with controls that preserve the component families while changing their functional organization. Under identical features and selection rules, naively merging the semantic prior and rationale teacher into one target is weaker than decoupled FDIL (55.66 **vs.** 57.49 **Avg F1**), although the strongest single-signal control remains close. We therefore claim a modest aggregate benefit together with separate measurability and deployability, not that decoupling is the only competitive design. In addition, shuffling rationales across images reduces Macro-F1 by 7.8 and Hard-F1 by 7.6, while removing or flattening the agreement gate has smaller effects. These controls show that the result depends on image-matched semantic content and its uncertainty-aware use rather than on adding arbitrary external text.

Revised manuscript: Sec. 3.5 and Eq. (15); Sec. 4, Tables 6, 13, and 10; Sec. 5.

4. Response to Reviewer 3 (Associate Editor)

We thank the Associate Editor for the focused, actionable points.

Comment 1

The authors are encouraged to clearly state, in the Abstract, the quantitative improvements achieved by the proposed FDIL method compared to existing approaches.

Response: Following this recommendation, we have revised the Abstract to report the controlled gains over representative recent methods: at least +3.4 **Avg F1**, +5.7 **Hard F1**, and +3.4 **threshold-free mAP** under matched CLIP ViT-L/14 features and validation-only thresholding.

Revised manuscript: Abstract (final sentence) and Table 2.

Comment 2

In Section 3.3.1, Semantic Prior Construction, the authors should provide a more detailed justification for the selected scenarios, including their representativeness and the expected outcomes in conditions not covered by the presented scenarios.

Response: We appreciate the request for an explicit selection criterion and uncovered-case behavior. The revised Semantic Prior Construction now clarifies that the scenarios are not hand-picked. A fixed generation protocol is applied uniformly to every intent class and enumerates visually distinct realizations along five axes (subjects/objects, actions, settings, atmosphere, and photographic style) with at least three settings per intent to span intra-class variation rather than a single canonical view. The objective is representative coverage, not an exhaustive list of every possible scene. Each source is normalized independently, and the scenario score is averaged with lexical and canonical scores before being used only to rerank the base model's Top-K candidates. Consequently, an image outside the generated scenarios does not trigger a brittle lookup failure: the scenario match weakens, while the lexical and canonical sources, unchanged base logits, and image-conditioned residual student continue to support the prediction. The prior-source ablation further shows that the three sources are complementary and that their ensemble provides the best aggregate coverage.

Revised manuscript: Sec. 3 (Semantic Prior Construction) and Table 7.

Comment 3

Table 2 shows that FDIL achieves lower performance than the conventional HLEG method on the Easy subset, yet this observation is not discussed. The authors should explain the possible reasons for this result and clarify whether certain characteristics of the Easy subset contribute to the similar performance between the two methods. Further discussion of the key observations in the table is recommended.

Response: This Easy-subset exception is now stated explicitly rather than absorbed into the aggregate result. FDIL scores 81.19, 0.30 below HLEG and slightly above the CLIP baseline (80.99). As discussed in the revised manuscript, Easy classes are largely visually unambiguous; a strong CLIP representation already separates them well, leaving little headroom for modules designed to resolve semantic overlap and uncertain supervision. The more diagnostic pattern appears on the Medium and Hard subsets: FDIL reaches 57.73 and 35.02, respectively, and its gains over the CLIP baseline are substantially larger there. Per-class attribution is consistent with this explanation: the largest improvements occur in abstract Hard classes such as *WorkILike*, *PassionAboutSomething*, and *SocialLifeFriendship*, while only one Easy class shows a small regression (-0.96 F1). We therefore present the near-tie on Easy as an expected scope characteristic of an ambiguity-targeted method, not as evidence of uniform dominance.

Revised manuscript: Sec. 4 (Performance on Difficulty- and Content-Dependent Subsets), Table 3; supporting per-class values in Table R1 of this letter.

5. Response to Reviewer 4

We thank the reviewer for the thorough review, which we address point by point.

Comment 1

The manuscript must require careful proofreading, such as [7] IEEE Trans. Affect. Comput., and so on.

Response: Thank you for catching this presentation issue. We performed a manuscript-wide proofreading pass and standardized bibliography venue names and abbreviations, including the flagged *IEEE Trans. Affect. Comput.* entry.

Revised manuscript: Bibliography and manuscript-wide language.

Comment 2

The paper is not very clear but only claims that " A central difficulty of this task lies in two complementary forms of ambiguity: supervisory ambiguity, caused by subjective annotation over plausible intentions, and semantic ambiguity, caused by semantically adjacent and overlapping intents." Please give more analysis about these disadvantages in the existing visual intention understanding.

Response: We appreciate this request for a sharper diagnosis. The Introduction and Related Work now explain that structure-aware and semantic-prior methods mainly address class-side overlap, whereas robust-loss and disagreement-aware methods mainly address supervision-side uncertainty; treating either as the sole source leaves the other failure mode unresolved. We also added empirical and theoretical support in the Method section. The candidate-recall diagnostic shows that the baseline Top-10 set already covers 83.9% of positive labels and retains 67.8% of false negatives as locally recoverable candidates, locating much of the remaining semantic error in local ranking and calibration. The accompanying excess-risk decomposition separates label-side conditional entropy from feature-side inter-class aliasing, explaining why uncertainty-guided supervision and semantic reranking address different factors of the risk. Thus, the two-ambiguity formulation is supported by a measurable diagnostic and a formal argument rather than asserted only as an architectural intuition.

Revised manuscript: Sec. 1, Sec. 2, and Sec. 3.5.

Comment 3

The clear purpose of work the proposed work is lacking in the introduction section.

Response: We have rewritten the Introduction to state the purpose directly: FDIL converts external semantic knowledge into a structured prior and an uncertainty-gated training signal that target semantic and supervisory ambiguity separately.

Revised manuscript: Sec. 1 (final related-work transition and contribution paragraphs).

Comment 4

The introduction should be re-discussed and extended. I think that more Large Language model-based visual intention understanding methods should be mentioned. In the introduction, this section is hard for readers to understand its latest development.

Response: To bring the literature discussion up to date, we have expanded it to include IntCLIP (ECCV 2024), IntentMLM (Pattern Recognition 2026), and the broader use of MLLMs as external semantic sources. Specifically, the revised text distinguishes semantic knowledge supplied by these models from the dataset-specific calibration and uncertainty-aware supervision required by FDIL, and links this distinction to the new zero-shot MLLM comparison.

Revised manuscript: Sec. 1, Sec. 2, and Table 2 (Zero-shot VLM group).

Comment 5

You may consider including a table of notations.

Response: We have followed this suggestion and added a compact notation table covering the principal symbols used in the method and theoretical analysis. To remain within the 35-page limit, the table focuses on the notation needed to follow the model formulation, ambiguity decomposition, and training objective rather than repeating symbols that are defined locally and used only once.

Revised manuscript: Sec. 3, Table 1.

Comment 6

There should be larger-scale datasets to better evaluate the proposed methods.

Response: We share the reviewer's view that evidence beyond Intentonomy is useful for assessing scope. We therefore added a controlled 5×5 cross-validation study on the multi-annotated EMOTIC validation and test pool (10,613 persons), using out-of-fold teacher predictions so that privileged distillation is evaluated without target leakage. Across 25 seed×fold runs, both standard and agreement-gated distillation improve the image-only baseline on all four aggregate metrics ($p < 0.01$ in the paired fold-level tests), while the gated variant is comparable to standard distillation overall.

The agreement-stratified results below provide the more informative view of the gate. On the lowest-agreement (1/3) subset, gated KD improves the image-only baseline by +0.43 **Macro-F1**, +0.69 **Micro-F1**, and +0.69 **Samples-F1** (all $p < 0.01$), and is numerically strongest among the two distillation variants on Macro-, Micro-, and Hard-F1. At agreement 2/3, it is strongest on Macro-F1, mAP, and Hard-F1. In contrast, no reliable gated gain appears on fully agreed samples, where the gate is designed to defer to direct supervision. The gated-versus-standard differences within individual slices are small and mostly not

Table R2: Secondary generalization evidence on the multi-annotated EMOTIC validation and test pool: controlled 5×5 cross-validation, mean over 25 seed×fold runs. $^{*}p < 0.01$, paired t-test vs. baseline.

Method	mAP	Macro-F1	Micro-F1	Samples-F1
Image-only baseline	42.57	43.34	60.32	60.01
+ standard KD	42.77*	43.68*	60.95*	60.75*
+ agreement-gated KD	42.77*	43.72*	60.97*	60.67*
Oracle teacher (privileged text)	66.43	60.96	74.63	75.68

Table R3: Agreement-stratified EMOTIC results over the same 25 seed×fold runs. Values are percentages; bold marks the best image-only student in each agreement group.

Agreement	Method	Macro	Micro	Samples	mAP	Hard
1/3	Baseline	44.15	62.19	62.03	44.15	36.92
	Standard KD	44.55	62.83	**62.75**	**44.37**	37.38
	Gated KD	**44.58**	**62.88**	62.72	44.32	**37.45**
2/3	Baseline	40.84	54.39	54.84	44.63	34.30
	Standard KD	41.04	**55.00**	**55.63**	44.85	34.44
	Gated KD	**41.20**	54.92	55.41	**45.18**	**34.69**
1	Baseline	22.42	36.31	36.60	34.07	**18.23**
	Standard KD	**22.83**	**37.28**	**37.80**	34.07	18.05
	Gated KD	22.15	36.79	37.20	**34.34**	17.41

significant, so we interpret this result conservatively: privileged semantic supervision transfers to EMOTIC, and agreement conditioning concentrates its useful behavior on uncertain samples, while Intentonomy remains the dataset that establishes the gate's incremental advantage with native training-time agreement. We have added this cross-dataset evidence and its limitations to the Discussion, showing that the framework has a degree of generalization beyond visual intention recognition without presenting EMOTIC as a new benchmark result.

Revised manuscript: Sec. 5 (Limitations and Future Directions); auxiliary results in Tables R2 and R3 of this letter.

Comment 7

Experimental evaluation is not sufficient. More comparisons with state-of-the-art Large Language Model methods (such as 2025 and 2026) and t-SNE are needed.

Response: In response, we have added zero-shot Qwen3-VL-8B-Instruct and InternVL3-8B under the same label set and output protocol, together with the published GPT-4o result from IntentMLM. Specifically, all three prompt-only models remain below the task-trained CLIP/FDIL models, so we treat MLLMs as useful semantic sources rather than substitutes for task-specific learning. We also added a t-SNE comparison on semantically adjacent intents, with the complete scores retained in the main-results table.

Revised manuscript: Sec. 4 (Zero-shot Multimodal Large Model Comparison and Interpretability Analysis), Table 2, and Fig. 3.

Comment 8

Please use some ROC-based curves for a clear evaluation of the proposed method with other state-of-the-art methods.

Response: We have added micro-averaged ROC and precision–recall curves for FDIL, the CLIP baseline, HLEG, LabCR, and IntCLIP using the same matched CLIP ViT-L/14 features and split. Specifically, FDIL is higher on both curves, and the accompanying table reports ROC-AUC, PR-AUC, mAP, confidence intervals, and paired tests. This keeps the visual comparison and the numerical evidence under the same controlled setting.

Revised manuscript: Sec. 4 (Calibration and Threshold-free Evaluation), Fig. 4, and Table 10.

Comment 9

A new section called “Discussion” could be added. The authors can discuss the main findings and also the important advantages and disadvantages of the proposed model.

Response: Following this suggestion, we have added a Discussion that summarizes the main findings and advantages and states the observed failure cases and generalization limits, including the *CreativeUnique* regression and the scope of the EMOTIC evidence.

Revised manuscript: Sec. 5.

Comment 10

It would add more value to the paper if more theoretical analysis were conducted, e.g., from information-theoretical or feature selection perspectives.

Response: We have added an information-theoretic excess-risk decomposition that separates label-side conditional entropy from feature-side inter-class aliasing. We also cast SLR-C as local feature selection and reranking: the candidate-recall diagnostic shows that most positives are already retrievable within the Top-10 set, so the remaining challenge is primarily to separate neighboring intents. The resulting predictions are evaluated by the component, negative-control, and unified-versus-decoupled experiments; the detailed control results are given in our response to Reviewer 2's novelty concern rather than repeated here.

Revised manuscript: Sec. 3.5, Eq. (15), and Sec. 4, Tables 6, 13, and 10.

Comment 11

A comparative analysis of model parameters, FLOPs, or inference times is necessary.

Response: We now quantify the deployment cost with a same-environment comparison of trainable parameters, FLOPs, and per-image latency for FDIL and the feature-level baselines, with all heads evaluated over the shared frozen CLIP features. Specifically, FDIL uses 2.41M parameters, 4.93M FLOPs, and 0.47ms per image; the table contains the full method-by-method comparison. IntentMLM is identified separately as an MLLM pipeline whose cost is not directly comparable to a feature-level head.

Revised manuscript: Sec. 4 (Efficiency Comparison), Table 15.

6. Response to Reviewer 5

We thank the reviewer for the positive assessment of the decoupling perspective and the module design, and for the precise minor corrections.

Comment 1

Figure 2 caption inconsistency: The caption refers to "(Top) UTD" and "(Bottom) SLR-C", but the figure shows "A" as the top and "B" as the bottom. Please revise the caption or the figure to ensure consistent correspondence.

Response: Thank you for identifying this inconsistency. To satisfy the page limit, we shortened the Figure 2 caption and removed the former "Top/Bottom" panel description. The revised caption now summarizes the functions of SLR-C and UTD without making the inconsistent correspondence, so this issue no longer appears in the manuscript.

Revised manuscript: Fig. 2 caption.

Comment 2

Missing experimental settings in Table 5:: Two additional configurations should be included: (i) Lexical + Scenario, and (ii) Canonical + Scenario.

Response: We have added both missing pairwise configurations for a complete ablation: Lexical+Scenario and Canonical+Scenario under the same K=10 reranking-only protocol. Specifically, both improve over their single-source components; Canonical+Scenario is the strongest pair on Hard-F1, while the three-source ensemble remains best on aggregate Avg-F1. The table reports the complete metric values.

Revised manuscript: Sec. 4 (Heterogeneous Priors in SLR-C), Table 7.

Comment 3

Unexplained anomaly in Table 6: Provide a justification for why $K = 5$ $K = 28$ yield better performance than $K = 10$.

Response: We revisited this anomaly by rerunning $K \in \{5, 10, 28\}$ over three seeds under the final protocol. The revised sweep identifies $K = 10$ as the best setting, so the earlier single-seed dip is not used in our conclusion. Specifically, $K = 5$ is slightly too restrictive and can exclude plausible positive labels, whereas $K = 28$ admits all classes and adds distractor noise without improving semantic coverage. A moderate $K = 10$ best balances candidate recall and local separability. We now use $K = 10$ throughout and have added this analysis directly to the sensitivity experiment accompanying the complete metrics table.

Revised manuscript: Sec. 4 (Sensitivity to Candidate Size K), Table 8.

Comment 4

Typo in Appendix tables: In Table B.1 and Table B.2 (page 35), “FDIR” should be corrected to “FDIL”.

Response: Thank you for identifying this typo. To meet the page limit, we removed the former Appendix Tables B.1–B.2 from the revised manuscript, so the reported instance no longer appears. We also performed a manuscript-wide terminology check and confirmed that every remaining occurrence uses the correct acronym “FDIL”.

Revised manuscript: Manuscript-wide terminology; former Appendix Tables B.1–B.2.

7. Response to Editor-in-Chief Checklist

We confirm that the revision adheres to the editorial checklist below.

Comment 1

Take a careful look at your bibliography and they cover the state of the art. Missing references from last and current year most probably would mean you are missing the state of the art and the revision process can be delayed being asked to update it. Please do not make excessive citation to arXiv papers, but substitute them with their peer-reviewed versions, or papers from a single conference series. Do not cite large groups of papers without individually commenting on them. So we discourage " In prior work [1,2,3,4,5,6] . . . ". Your bibliography in the final version after the revision still should be between 35–55 items.

Response: Following this guidance, we carefully reviewed and updated the bibliography. The revision now includes recent peer-reviewed work such as IntCLIP (ECCV 2024) and IntentMLM (Pattern Recognition 2026), replaces preprints where published versions are available, and discusses the most relevant methods individually. The bibliography now contains **40 references**, within the requested range.

Revised manuscript: Sec. 1, Sec. 2, and Bibliography.

Comment 2

Please make sure the revised version is relevant to the readership of the Pattern Recognition field. To this end, please make sure you cite RECENT work from the field of pattern recognition not only the Pattern Recognition journal.

Response: We have strengthened the connection to the broader pattern-recognition community in the Introduction and Related Work, including recent visual intent recognition, uncertainty learning, CLIP-based recognition, and multimodal large-model research.

Revised manuscript: Sec. 1, Sec. 2, and Bibliography.

Comment 3

Although the revision could lead to extending your article, it still can not exceed the page limits or violate the format, i.e. double spaced SINGLE column with a maximum of 35 pages for a regular paper and 40 pages for a review.

Response: The revised article now follows the required double-spaced, single-column format and is exactly **35 pages**. Specifically, we placed the compact notation table into the Method section, combined related interpretability and UTD analyses, removed redundant validation-split tables, and placed exact generation prompts in the artifact release. No reviewer-requested conclusion was removed; detailed auxiliary results are summarized in this letter or released with the artifacts.

Revised manuscript: Full revised manuscript (double-spaced, single-column, 35 pages); Table 1; Data and artifact availability statement.

8. Closing Statement

We believe the revision substantially strengthens the novelty framing, the fairness and interpretability of the evaluation, and the reproducibility of the work. We thank the Associate Editor and reviewers again for their careful guidance and hope that the revised manuscript now addresses all concerns.